Predicting Spotify Track Popularity through Bayesian Reasoning

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Abstract

This mini-project applies Bayesian Networks to the problem of understanding what drives the popularity of a music track. We build a small, custom-defined Discrete Bayesian Network over seven variables derived from the Spotify Tracks Dataset (genre, tempo, danceability, energy, valence, explicit content, and popularity), fit its parameters via Maximum Likelihood Estimation, and use exact inference to answer three concrete questions about track popularity. We find that energy is a considerably stronger driver of predicted popularity than mood (valence), and that genre alone produces large shifts in the odds of a track being popular.

Introduction

Domain

Streaming platforms such as Spotify expose, for each track, audio features computed from the signal itself (danceability, energy, valence, tempo, etc.) together with a popularity score. This project models the probabilistic relationships between a track's genre, tempo, audio characteristics, and resulting popularity, using the Spotify Tracks Dataset (Pandya 2023).

Aim

The purpose of this project is to build a Bayesian Network relating a track's genre, tempo and audio characteristics to its popularity, and to use it to answer the following questions:

- Does genre alone shift the odds of a track being popular?
- Does a track's mood (valence) matter more than its energy for predicting popularity?
- Do audio features depend on each other beyond what genre alone explains (e.g. does tempo drive energy, which drives mood)?
- How large a swing in predicted popularity can be produced by combining favorable vs. unfavorable evidence across all features?

Method

We model the problem with a **Discrete Bayesian Network**: a directed acyclic graph (DAG) in which each node is a categorical random variable (a finite number of possible states, no continuous values) and each node's distribution is conditioned only on its parents in the graph. We used the pgmpy library to define this DAG manually (structure not learned from data) and to fit its Conditional Probability Tables (CPTs) via Maximum Likelihood Estimation. We then used Variable Elimination to perform exact inference and answer the questions above.

Results

The clearest single driver is energy, not mood or genre in isolation. Classical and jazz stand out as weaker performers; combining several favorable features at once produces a much larger swing in predicted popularity than any single feature alone.

Model

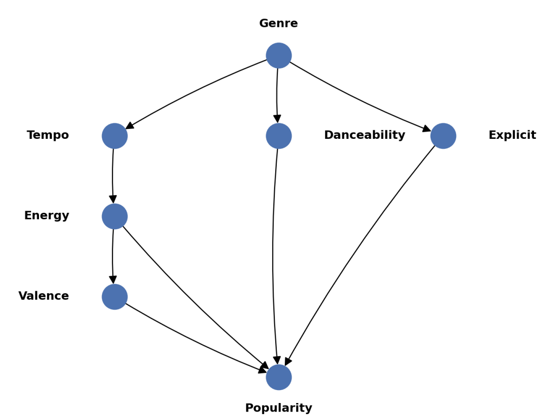

Figure 1: The custom Bayesian Network relating genre and audio features to track popularity.

The dataset was restricted to 6000 tracks evenly split across six genres (pop, classical, jazz, metal, edm, hip-hop), chosen to span a wide range of average popularity and musical style while keeping the model small and the CPTs well estimated. Continuous audio features (danceability, energy, valence, tempo) and the popularity score were discretized into three bins each using fixed range cuts (rather than quantile-based cuts, since popularity is heavily skewed towards zero and quantile binning produced degenerate bins). The seven nodes are:

- **Genre**: one of the six chosen genres.
- **Tempo**: slow/medium/fast (bpm-based).
- **Danceability**, **Energy**: low/med/high, derived from Spotify's own 0–1 audio feature scores.
- **Valence**: sad/neutral/happy, Spotify's measure of musical mood/positiveness.
- **Explicit**: boolean flag for explicit content.
- **Popularity**: low/med/high (target node).

A structure where Genre is the sole cause of every audio feature, with all features feeding directly into Popularity, would assume that once Genre is known, the audio features are mutually independent – unrealistic: slow tracks are low-energy 47% of the time vs. 19% for fast tracks, and low-energy tracks are "sad" 51% of the time vs. 13% "happy". We therefore modeled these dependencies explicitly with **Genre** $\rightarrow$ **Tempo** $\rightarrow$ **Energy** $\rightarrow$ **Valence**, alongside Genre's direct influence on Danceability and Explicit, with all four audio/content features feeding into Popularity. Popularity's parent set has only these four direct parents, keeping its CPT size and data-sufficiency manageable (47/54 parent combinations present, minimum count 1); the two additional CPTs (Energy$|$Tempo, Valence$|$Energy) are fully populated (9/9 combinations each, minimum count $>$175).

Analysis

Experimental setup

We ran four sets of exact-inference queries using Variable Elimination: (1) $P(\text{Popularity} \mid \text{Genre} = g)$, where g ranges over the six genres, with no other evidence; (2) $P(\text{Popularity} = \text{high} \mid \text{Valence} = v)$ and $P(\text{Popularity} = \text{high} \mid \text{Energy} = e)$, where $v \in \{\text{sad}, \text{neutral}, \text{happy}\}$ and $e \in \{\text{low}, \text{med}, \text{high}\}$, to compare the two features' individual effect on popularity; (3) a best-case scenario (pop, high danceability, high energy, happy, non-explicit) versus a worst-case scenario (classical, low danceability, low energy, sad, non-explicit); (4) $P(\text{Popularity} = \text{high} \mid \text{Tempo} = t)$, where $t \in \{\text{slow}, \text{medium}, \text{fast}\}$, testing whether tempo shifts popularity odds through its effect on energy.

Results

Genre. Classical (50.6% low popularity, 17.8% high) is the weakest genre for popularity, while hip-hop/edm/metal do best (24–28% high) – part of genre's effect is mediated through the Tempo$\rightarrow$Energy$\rightarrow$Valence chain rather than acting directly on every feature.

Valence vs. energy. Energy shows a clear, monotonic effect: P(high popularity) rises from 12.6% (low) to 24.8% (medium) to 27.9% (high). Valence has a small, non-monotonic effect (25.3% / 24.7% / 18.3% for sad / neutral / happy) – energy is clearly the dominant driver of the two.

Tempo. Slow tracks are noticeably less likely to be highly popular (18.0%) than medium (24.7%) or fast (24.3%) tracks, consistent with slow tracks tending toward lower energy – itself the strongest direct driver of popularity. Tempo's own effect on popularity, in other words, runs mainly through energy, matching what the structure assumes.

Combined scenarios. Best-case yields 21.7% chance of high popularity (43.7% low, 34.6% med) vs. 2.5% for worst-case (88.8% low) – essentially unchanged from a Genre-only structure, since these scenarios directly fix Danceability/Energy/Valence.

Diagnostic reasoning. We also ran a diagnostic (bottom-up) query, $P(\text{Genre} \mid \text{Popularity})$: given high popularity, classical drops to 12.7% (vs. 18.4% given low popularity), while hip-hop rises to 20.0% (vs. 15.5%) – confirming the network reasons correctly from effect back to cause.

Popularity's parent set – Danceability, Energy, Valence, Explicit – implies Genre is d-separated from Popularity given these four features (genre affects popularity only *through* the audio features it tends to produce, not directly), and the Tempo$\rightarrow$Energy$\rightarrow$ Valence chain implies Genre is d-separated from Valence given Energy alone. We verified both hold via exact reachability analysis; both are structural properties of the DAG we chose, reflecting modeling assumptions, not discovered facts.

Conclusion

The main takeaway is that lifestyle/production factors captured by audio features – particularly energy – outweigh a track's genre label as a driver of predicted popularity, and that modeling dependencies *among* those features (rather than treating them as independent given genre) changes which pathways carry that influence, without costing us any data-sufficiency for the target node. The main limitations are the discretization of continuous features into only three bins each; the restriction to six genres out of 114, for tractability; the network structure being chosen from domain intuition and confirmed correlations rather than formal structure-learning or independence testing; Maximum Likelihood Estimation falling back to a uniform distribution for the few parent combinations absent from our sample, rather than a Dirichlet-smoothed estimate; and the source dataset's exactly-1000-per-genre sampling, which gives Popularity's Genre prior an artificial uniformity not reflective of real-world catalog proportions.

Links to external resources

- Dataset: `https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset`
- Code repository: `https://github.com/MattiaSemproli/Spotify-Bayesian-Network.git`

References

Pandya, M. 2023. Spotify tracks dataset. Kaggle.